



YOUCHENG ZONG

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🏛 Northeastern University

I work on data-driven and LLM-assisted intelligence for complex industrial processes, with a particular focus on metallurgy, process decision support, adaptive prediction, and industrial time-series modeling.

🎓 Education

2024.09-Present	Northeastern University Ph.D. Student in Control Science and Engineering	Doctor of Philosophy
2021.09-2024.06	Anhui University of Technology Master's Degree in Materials and Chemical Engineering	Master of Engineering
2017.09-2021.06	Anhui University of Technology Bachelor's Degree in Automation	Bachelor of Engineering

📄 Publications

- › LLM-driven human-AI collaborative decision support system for complex industrial processes: A case study in metallurgy.
Youcheng Zong, Runda Jia, Kang Li, Dazhan Xue, Liqiang Zhang, Dakuo He.
Neural Networks (NN), 2026. [DOI](#).
- › A meta-contrastive learning hybrid model for adaptive temperature trend prediction in variable ladle preheating.
Youcheng Zong, Runda Jia, Shuai Wu, Liqiang Zhang, Dakuo He.
Engineering Applications of Artificial Intelligence (EAAI), 2025. [DOI](#).
- › Hybrid grid search and Bayesian optimization-based random forest regression for predicting material compression pressure in manufacturing processes.
Youcheng Zong, Yi Nian, Chaojie Zhang, Xinyu Tang, Lin Wang, Liqiang Zhang.
Engineering Applications of Artificial Intelligence (EAAI), 2025. [DOI](#).
- › Iron-tapping state recognition of blast furnace based on Bi-GRU composite model and post-processing classifier.
Youcheng Zong, Shaowei Hu, Debo Qin, Zixian Wang, Chaojie Zhang, Jianhua Chu, Liqiang Zhang.
IEEE Sensors Journal (ISJ), 2023. [DOI](#).
- › Judgment of blast furnace iron-tapping status based on data differential processing and dynamic window analysis algorithm.
Youcheng Zong, Zixian Wang, Xiang Liu, Yi Nian, Jianian Pan, Chaojie Zhang, Yingxue Wang, Jianhua Chu, Liqiang Zhang.
Progress in Natural Science: Materials International (PNSMI), 2023. [DOI](#).
- › Zero-shot capillary segmentation in dermoscopy images via SAM2: A case study on oral mucosa.
Weilan Su, **Youcheng Zong**, Runda Jia, Jian Qin, Ming Li.
IEEE Journal of Biomedical and Health Informatics (JBHI), 2025. [DOI](#).
- › Influence of heavy reduction during solidification process of billets based on 3D reconstruction of dendrites.
Yi Nian, **Youcheng Zong**, Chaojie Zhang, Xinyu Tang, Jiale Li, Liqiang Zhang.
Journal of Iron and Steel Research International (JISRI), 2025. [DOI](#).

- Application of electromagnetic metallurgy in continuous casting: A review.
Jiale Li, Yi Nian, Xiang Liu, **Youcheng Zong**, Xinyu Tang, Chaojie Zhang, Liqiang Zhang.
Progress in Natural Science: Materials International (PNSMI), 2024. DOI.
- XGBoost and CNN-based prediction of external forces and stress-strain during solidification reduction in casting.
Yi Nian, Chaojie Zhang, Xinyu Tang, **Youcheng Zong**, Jiale Li, Liqiang Zhang, Yingxue Wang.
Progress in Natural Science: Materials International (PNSMI), 2025. DOI.

Granted Patents

- First inventor. A method for judging the iron-tapping state of blast furnace hot-metal ladles. Patent No. TBD.
- First inventor. A pressure prediction method and system for high-performance material compression processes. Patent No. TBD.
- Third inventor. A device and method for synchronous detection of reduction amount and pressure value in continuous-casting soft reduction. Patent No. TBD.
- First inventor. An intelligent temperature and humidity regulation system for poultry incubation based on a mobile app. Patent No. TBD.

Software Copyrights

- First owner. Ten software copyrights, including systems for continuous-casting billet weighing monitoring, intelligent scrap batching, and team collaboration platforms. Registration Nos. TBD.

Project Experience

- Industry-university-research project on intelligent transformation of a Baowu Changgang plant; responsible for equipment selection, communication-network construction, algorithm design, and software development.
- Industry-university-research project on process-control efficiency improvement for a Baowu Magang LF furnace process; responsible for equipment selection, communication-network construction, algorithm design, and software development.
- General Program of the National Natural Science Foundation of China; participated in data-driven mechanism analysis for a rheological-process-related research topic.
- Young Scientists Fund of the National Natural Science Foundation of China; participated in data-driven mechanism analysis for large-ingot solidification research.
- National Key Research and Development Program project; participated in endpoint-prediction software development for a 5G-enabled beneficiation and smelting process.



宗有成

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东北大学

本人主要研究复杂工业过程中的数据驱动与大模型辅助智能, 重点关注冶金过程、过程决策支持、自适应预测和工业时间序列建模。

教育经历

2024.09-至今	东北大学	博士研究生
	控制科学与工程专业博士研究生	
2021.09-2024.06	安徽工业大学	工程硕士
	材料与化工专业硕士	
2017.09-2021.06	安徽工业大学	工学学士
	自动化专业学士	

论文发表

- LLM-driven human-AI collaborative decision support system for complex industrial processes: A case study in metal-lurgy.
Youcheng Zong, Runda Jia, Kang Li, Dazhan Xue, Liqiang Zhang, Dakuo He.
Neural Networks (NN), 2026. DOI.
- A meta-contrastive learning hybrid model for adaptive temperature trend prediction in variable ladle preheating.
Youcheng Zong, Runda Jia, Shuai Wu, Liqiang Zhang, Dakuo He.
Engineering Applications of Artificial Intelligence (EAAI), 2025. DOI.
- Hybrid grid search and Bayesian optimization-based random forest regression for predicting material compression pres-sure in manufacturing processes.
Youcheng Zong, Yi Nian, Chaojie Zhang, Xinyu Tang, Lin Wang, Liqiang Zhang.
Engineering Applications of Artificial Intelligence (EAAI), 2025. DOI.
- Iron-tapping state recognition of blast furnace based on Bi-GRU composite model and post-processing classifier.
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- Judgment of blast furnace iron-tapping status based on data differential processing and dynamic window analysis algo-rithm.
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Progress in Natural Science: Materials International (PNSMI), 2023. DOI.
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- › XGBoost and CNN-based prediction of external forces and stress-strain during solidification reduction in casting.
Yi Nian, Chaojie Zhang, Xinyu Tang, **Youcheng Zong**, Jiale Li, Liqiang Zhang, Yingxue Wang.
Progress in Natural Science: Materials International (PNSMI), 2025. DOI.

✿ 授权专利

- › 第一发明人。一种高炉铁水包出铁状态的判断方法。
- › 第一发明人。一种高性能材料压缩过程的压力预测方法及系统。
- › 第三发明人。一种连铸轻压下过程压下量及压力值同频检测装置与方法。
- › 第一发明人。一种基于移动 APP 的家禽孵化智能温湿度调节系统。

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🔧 项目经历

- › 产学研项目, 宝武长钢某厂智能化改造, 负责设备选型、通讯网络搭建、算法设计和软件开发。
- › 产学研项目, 宝武马钢 LF 炉某流程控制增效, 负责设备选型、通讯网络搭建、算法设计和软件开发。
- › 国家自然科学基金面上项目, 参与流变过程相关研究中的数据驱动机理分析。
- › 国家自然科学基金青年科学基金项目, 参与大型钢锭凝固过程研究中的数据驱动机理分析。
- › 国家重点研发计划课题级项目, 参与基于 5G 的选冶过程终点预测软件开发。